

NAAMII

Jwagal, Lalitpur, Nepal

Brief Report on

Task 1

Identify the medial and lateral lowest points on the tibial surface

Submitted by

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Femur and Tibia Segmentation and Landmark Detection: Report

1. Introduction

This project automates the segmentation of femur and tibia bones from 3D CT scans, as well as identifies anatomical landmarks on the tibial surface. The tasks were completed using classical image processing techniques and Python-based medical imaging libraries. The code and documentation are available on GitHub at:

<https://github.com/PrabeshTimilsina/NAAMII-task-1>

2. Approach and Methodology

2.1. Segmentation Pipeline

- **Denoising:**
Gaussian filtering was applied to the CT images to reduce noise and enable thresholding.
- **Bone Segmentation:**
A fixed Hounsfield Unit (HU) threshold was used to segment bone tissue from the CT volume.
- **Mask Cleaning:**
Morphological operations like hole filling and removal of small objects were used to clean the binary mask.
- **Bone Separation:**
Connected component analysis was performed to separate the femur and tibia, assigning the correct label based on anatomical position.
- **Mask Saving:**
The resulting femur and tibia masks were saved in `.nii.gz` format for further processing.

2.2. Mask Expansion and Randomization

- **Expansion:**

The tibia mask was expanded by 2 mm and 4 mm using distance map thresholding, maintaining the expanded mask anatomically reasonable.

- **Randomized Masks:**

Two additional masks were generated by adding random expansion thresholds, which lay between the original and maximally expanded masks.

2.3. Landmark Detection

- **Surface Extraction:**

Surface voxels were identified using morphological operations.

- **Medial and Lateral Lowest Points:**

For each mask, the medial and lateral lowest points on the tibial surface were found by separating surface voxels with the inner voxel along the medial-lateral axis and selecting the lowest point on each side.

- **Coordinate Conversion:**

Voxel coordinates were converted to real-world coordinates using the image origin, spacing, and direction metadata.

3. Use of AI Tools

As I had limited prior experience with medical image analysis, I used Perplexity AI to:

- Research best practices and standard approaches for bone segmentation and landmark detection.
- Clarify concepts related to medical image metadata, anatomical orientation, and general medical image analysis.
- Refine and revise code segments for robust and reproducible solutions.
- Prepare technical documentation, Readme file, and this report to industry standards.

I wrote, reviewed, and tested all code by myself, using Perplexity AI as an interactive reference and coding assistant.

4. Results

- All segmentation masks needed (original, 2mm expanded, 4mm expanded, randomized 1, randomized 2) were generated and saved in `.nii.gz` format.
- Medial and lateral lowest tibial surface points were computed for each mask and saved in a CSV file for submission.
- The codebase is modular, well-documented, and follows best practices for reproducibility and clarity.

5. Conclusion

This project demonstrates a complete classic pipeline for bone segmentation and anatomical landmark detection from CT scans. The approach is robust, reproducible, and open to further extension or integration into clinical or research pipelines.

6. References

- SimpleITK Documentation: <https://simpleitk.readthedocs.io/>
- scikit-image Documentation: <https://scikit-image.org/>
- Perplexity AI: <https://www.perplexity.ai/>